Introduction to Transfer Learning

Thursday, 5 November 2026 9:51 AM

Agenda

- Recognition of world famous landmark
- Standard CNN architecture like AlexNet and VGGNet
- Transfer learning using VGGNet

* Top-K Accuracy

True label	Top-3 Predicted labels	
Cat	**Cat**, Lion, Dog	✓
Dog	Giraffe, Lion, Cat	×
Lion	Cat, **Lion**, Dog	✓
Giraffe	**Giraffe**, Dog, Cat	✓
Dolphin	**Dolphin**, Cat, Giraffe	✓

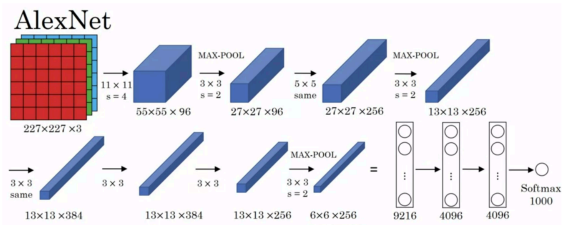
An input image of shape (227, 227, 3) is convolved with an (11, 11) kernel using 96 filters (stride 1, no padding), giving an output of shape (217, 217, 96). What is the total number of parameters?

$11 \times 11 \times 3 = 363 \times 96 = 34848 + 96 = 34,944$ (96: bias)

$(11 \times 11 \times 3 + 1) * 96 = 34,944$ (1: bias)

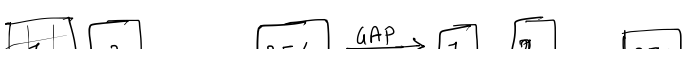
$\frac{N - k + 2x0}{4} + 1 = \frac{227 - 11}{4} + 1$

$= \frac{216}{4} + 1 = 55$



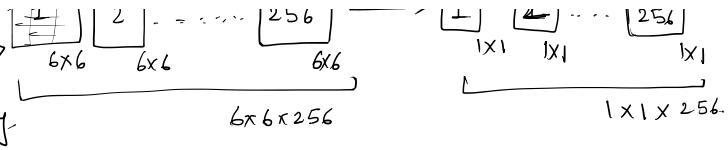
Layer	Filter/neurons	Filter size	Stride	Padding	Size of feature map
Input					227 x 227 x3
Conv1	96	11x11	4		55 x 55 x 96
Max pool1		3 x 3	2		27 x 27 x 96
Conv2	256	5 x 5	1	2	27 x 27 x256

Feature Extraction



from Images

Max Pool2		3 x 3	2		13 x 13 x 256
Conv 3	384	3 x 3	1	1	13 x 13 x 384
Conv 4	384	3 x 3	1	1	13 x 13 x 384
Conv 5	256	3 x 3	1	1	13 x 13 x 256
Max Pool 3		3 x 3	2		6 x 6 x 256
Dropout 1	Rate = 0.5				6 x 6 x 256
FC 1					4096
Dropout 2	Rate = 0.5				4096
FC2					4096
FC3					1000



Flattening

of classes.

Figure : AlexNet layers Computation

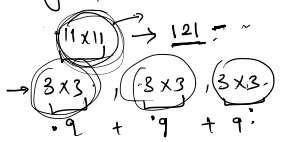
AlexNet with 5 Conv layers and 3 FC layers has ≈ 61 M parameters.

Fully connected.

* Drawbacks of AlexNet -

1) We do not use Batch Norm.

2) Too many parameters → 11 X 11 Kernel.

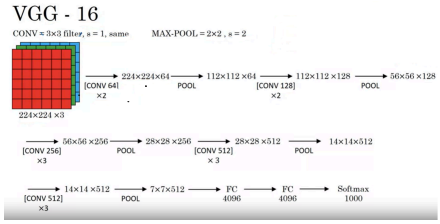


3) GAP is not used

* VGG - 16 / 19

of layers.

Total # of params in VGG16 → ≈ 15 M.

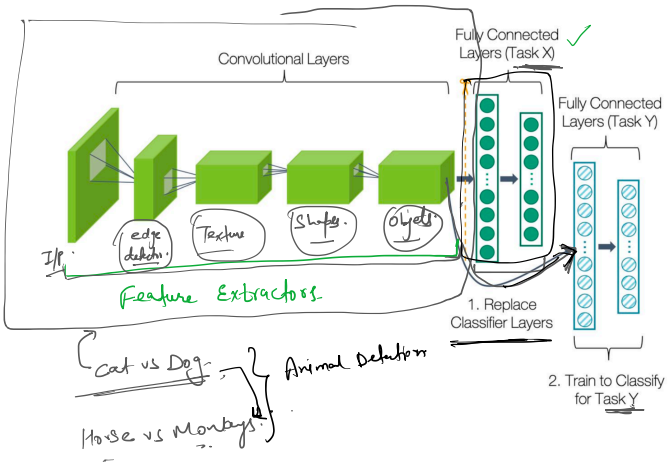


	Layer	Filter/neurons	Filter size	Stride	Padding	Size of feature map
	Input					224x224x3
1	Conv	64	3x3	1	same	224x224x64
2	Conv	64	3x3	1	same	224x224x64
	Max Pool		2x2	2		112x112x64
3	Conv	128	3x3	1	same	112x112x128
4	Conv	128	3x3	1	same	112x112x128
	Max Pool		2x2	2		56x56x128
5	Conv	256	3x3	1	same	56x56x256
6	Conv	256	3x3	1	same	56x56x256
7	Conv	256	3x3	1	same	56x56x256
	Max Pool		2x2	2		28x28x256
8	Conv	512	3x3	1	same	28x28x512
9	Conv	512	3x3	1	same	28x28x512
10	Conv	512	3x3	1	same	28x28x512
	Max Pool		2x2	2		14x14x512
11	Conv	512	3x3	1	same	14x14x512
12	Conv	512	3x3	1	same	14x14x512
13	Conv	512	3x3	1	same	14x14x512
	Max Pool		2x2	2		7x7x512
14	FC					4096
15	FC					4096
16	FC					1000 → classes

Figure: VGG16 layers Computation

Kernels → (3,3), (2,2)

↑ Conv ↑ Maxpool


Convolutional Layers
Fully Connected Layers (Task X)
Fully Connected Layers (Task Y)
edge detection
Texture
Shapes
Objects
I/p
Feature Extractors
1. Replace Classifier Layers
2. Train to Classify for Task Y
Cat vs Dog
Horse vs Monkeys
Animal Detection